

hedge for both expected and unexpected inflation. Fama and Schwert, however, used the housing component in the CPI, as they lacked direct real estate return data. For direct real estate, the results seem to be inconclusive. On the one hand, Goetzmann and Valaitis (2006) argue that direct real estate is a good inflation hedge. On the other hand, Huang and Hudson-Wilson (2007) find “that the conventional wisdom that real estate, as an asset class, is an effective inflation hedge is overly generous.” It is clear from the literature that REITs do not hedge inflation in the short run.⁴¹ Gyourko and Linneman (1998), for example, note that REITs are perverse inflation hedges.

I believe real estate offers some inflation protection but is far from an effective inflation hedge. That is, real estate is only partially real.

In Figure 11.13, I plot correlations of direct and indirect real estate returns with inflation over various horizons. I also overlay correlations of equity returns with inflation for comparison. We saw in chapter 8 that equity was a poor inflation hedge. Panel A plots classical (Pearson) correlations. REIT return correlations with inflation are slightly higher than equity correlations, but only just, and these increase from just above 0.1 at the one-quarter horizon to around 0.4 at the five-year horizon. NCREIF correlations are around 0.4 at the one-quarter horizon and decrease slightly to 0.3 at the five-year horizon. NCREIF returns, as I emphasized before, should not be trusted at short horizons due to illiquidity biases. It is comforting, though, that at long horizons, where the illiquidity biases are mitigated, the correlations for both REIT and NCREIF returns are both around 0.3 to 0.4. Real estate provides a partial inflation hedge at long horizons but not at short horizons.

Panel A of Figure 11.13 is an overly generous representation. In Panel B, I graph robust (Spearman) correlations. These are lower than the Pearson correlations in Panel A. Both the REIT and stock correlations with inflation are negative at short horizons, with correlations around -0.1 . (This is consistent with the findings for equities in chapter 8.) After one year, REIT correlations with inflation are a little higher than stock correlations. At the five-year horizon, the correlation of REIT returns with inflation is approximately 0.4, very similar to the number in Panel A. The robust calculations in Panel B for the NCREIF series have brought down the correlations with inflation significantly. Now the NCREIF correlations with inflation are around 0.2 at the one-quarter horizon, have a hump-shaped pattern increasing to around 0.3, and end at the five-year horizon back around 0.2.

In Panel C of Figure 11.13, I graph correlations of inflation with a common real estate factor from June 1980 to December 2011. I construct this real estate factor

⁴¹ Glascock, Lu, and So (2002) argue that this is because the observed negative relation between REIT returns and inflation is statistically spurious and explained by other macro factors, especially monetary policy shocks. This is similar to Fama’s (1981) argument on why equities are a poor hedge to inflation (see chapter 8). Case and Wachter (2011) take a contrary view and argue that REITs are good inflation hedges, but they do not define inflation-hedging ability as a correlation (or beta) with inflation.